

check_cm_healthcheck_covers_served_ports

scripts/pre_build/ check_cm_healthcheck_covers_served_ports.sh — CM- HEALTHCHECK-COVERS-SERVED- PORTS

Revision: 1 **Last modified:** 2026-08-20T17:30:00Z **Status:** active **Item:** BOB-138 (defect fixed) / BOB-140 (engine upstreamed per §11.4.177)

Overview

Asserts that every port a compose service SERVES appears in that service's health check. A check probing only a subset asserts a PROXY signal ("one port answers") in place of the REAL condition ("this service is serving"), so a service can report healthy indefinitely while its primary capability is dead (§11.4.201).

Wired as **pre-build invariant 44**, BLOCKING.

Why it exists — the forensic anchor (BOB-138, measured 2026-08-20)

The download-proxy service serves 7186 and 7187 from ONE process, but its health check probed only 7186. On a container reporting Up 4 hours (healthy):

```
curl --max-time 6 http://localhost:7186/      -> HTTP 200 in 0.096s
curl --max-time 6 http://localhost:7187/health -> HTTP 000 after 6.004s
```

The merge search service — the product’s primary capability — had been dead for roughly two hours, and nothing in the stack could observe it. An operator, a restart policy, and any depends_on: service_healthy all read “healthy”.

Prerequisites

- bash
- python3 with **PyYAML**. If no interpreter with PyYAML is reachable the gate **FAILs** — it does not SKIP. A gate that cannot parse cannot assert, and a quiet zero from a blind instrument is indistinguishable from a clean tree (§11.4.201(6)).
- The constitution submodule checked out (the detection engine lives there).

Usage

```
bash scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh
bash scripts/pre_build/
    check_cm_healthcheck_covers_served_ports.sh --help
bash scripts/pre_build/check_cm_healthcheck_covers_served_ports.sh
    COMPOSE MANIFEST
```

Exit codes: 0 PASS · 1 FAIL · 2 engine missing or engine usage error.
Invariant 44 treats any non-zero (other than a timeout) as FAIL, so a missing submodule blocks rather than silently passes.

Internal behaviour

This file is a **thin delegator** (§11.4.177 / §11.4.28): it holds only boba’s scope DATA and execs the shared engine at constitution/scripts/gates/cm_healthcheck_covers_served_ports.sh. No detection logic lives here, so it cannot drift from the engine. The engine carries no project literal; both the compose file and the manifest are inputs.

The engine reports a finding when a service:

- serves a declared port its health check does not probe;
- serves ports but declares no health check at all;

- is declared in the manifest but absent from the compose file (stale entry);
- publishes ports or declares a health check but is **UNDECLARED** in the manifest — silence is not an exemption.

Ports match as whole numbers, so 7187 never matches inside 71870 or 17187.

The served-port manifest is DATA, and why it is declared not derived

config/served_ports.yaml (§11.4.35) lists, per compose service, the ports it serves. These services use `network_mode: host`, so there is no ports: mapping to derive from, and the environment mixes two indistinguishable meanings:

```
PROXY_PORT=7186          <- a port this service SERVES
MERGE_SERVICE_PORT=7187  <- a port this service SERVES
QBITTORRENT_PORT=7185    <- a port this service CONNECTS TO
```

A gate sweeping `*_PORT` would demand a health check for a port the service does not serve — a false-positive refusal, which §11.4.201(1) forbids exactly as firmly as a false pass.

This trap is not hypothetical. The manifest's first version declared `qbittorrent-proxy-go: serves [7186, 7187, 7188]`, taken from CLAUDE.md prose. The Dockerfile runs one binary binding 7187, so the gate failed a service whose health check was already correct. The manifest is now derived from source, and the doc was corrected (BOB-141).

Edge cases

- **Zero services checked** → FAIL, not PASS. The gate refuses to call a blind run clean.
- **Both test: spellings** (list form and string form) are handled.
- **Compose parsing limits** (pre-existing, unchanged): a single compose file, with no extends, overlay or profile resolution. Not currently exercised by boba, but a consumer using overlays should know the manifest is checked against the base file only.

Related

- Engine: constitution/scripts/gates/cm_healthcheck_covers_served_ports.sh
- Paired §1.1 mutation test: constitution/scripts/gates/cm_healthcheck_covers_served_ports_mutation_test.sh (13 fixtures: 11 FAIL-on-mutation, 2 PASS-on-clean negative controls)
- Data: config/served_ports.yaml
- Evidence: docs/qa/B0B-138/closure-evidence.md, docs/qa/B0B-140/closure-evidence.md
- Wiring: scripts/pre_build_verification.sh invariant 44

Last verified: 2026-08-20 — delegator PASS (5 services verified), mutation test META PASS 13/13, both fail-closed properties confirmed through the delegator.